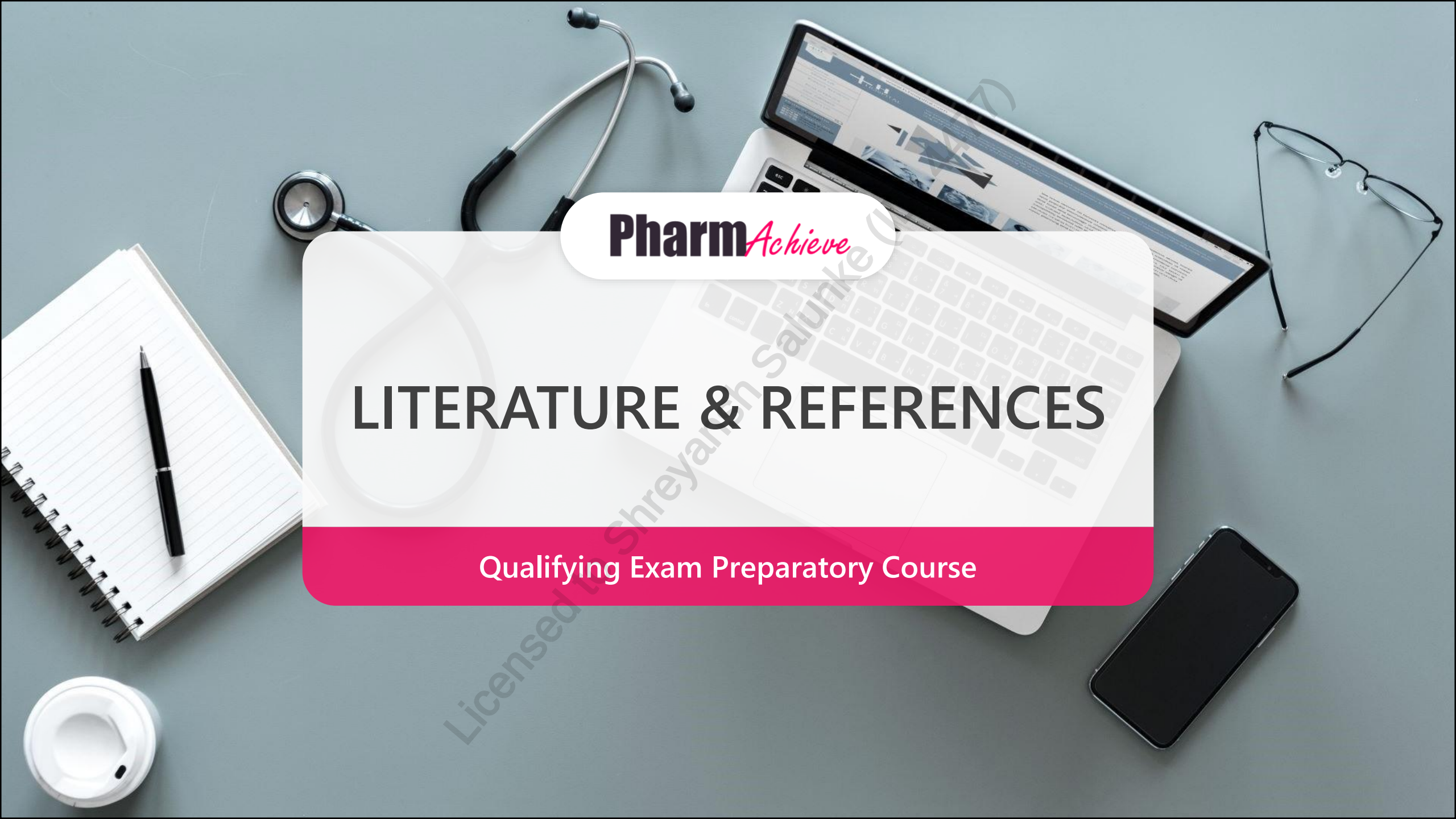




Pharm*Achieve*

LITERATURE & REFERENCES

Qualifying Exam Preparatory Course

COMPETENCIES COVERED IN THIS PRESENTATION...

Providing Care:
Clinical Care

Providing Care:
Distribution

Knowledge
& Expertise

Communication
& Collaboration

Leadership &
Stewardship

Professionalism

LEGEND

AD	Academic Detailing
AHFS	American Hospital Formulary Service
CE	Continuing Education
CPhA	Canadian Pharmacists Association
CTC	Compendium of Therapeutic Choices
CTMA	Compendium Of Therapeutics For Minor Ailments
CPS	Compendium Of Pharmaceuticals And Specialties
DIRC	Drug Information And Research Centre
FDA	Food and Drug Administration
ISMP	Institute For Safe Medication Practices
IV	Intravenous
MeSH	Medical Subject Headings
NAPRA	National Association Of Pharmacy Regulatory Authorities
NHP	Natural Health Product

NLM	National Library Of Medicine
OCP	Ontario College Of Pharmacists
PEBC	Pharmacy Examining Board of Canada
TdP	Torsades de Pointes

LITERATURE AND REFERENCES: TYPES OF REFERENCES

PRIMARY

- **ORIGINAL RESEARCH** authored by researchers and are often **published in peer-reviewed journals**
- Also include conference proceedings, technical reports
- Latest and current information
- Controlled clinical trials, case-control trials, case reports, cohort studies (e.g. Canadian Pharmacists Journal, New England Journal of Medicine)

SECONDARY

- Includes works that **build upon original research** including:
 - **Databases** that index or allow you to search primary literature (e.g. MEDLINE/PubMed, Web of Science, Google Scholar)
 - **Meta-analyses** that review original research (e.g. **Cochrane Reviews**)
 - **Practice Guidelines** that review the research (e.g. Diabetes Guidelines)

TERTIARY

- **Distillation and collection of primary and secondary sources**
- Room for bias
- Example: Textbooks, CPS, UpToDate® Lexidrug™, Remington, Martindale, etc.

MINIMUM LIBRARY REQUIREMENTS

The following are Ontario's requirements for what every Pharmacy **must have** in their library

1. Canadian Drug Reference / Compendium: CPS: Drug Information
2. Drug Interaction Publication: Micromedex, Lexi-Interact
3. Drug Therapy Publication: CPS: Therapeutic Choices, CPS: Minor Ailments
4. Patient Counselling Guide: Micromedex Care Notes
5. Setting-Specific References / Additional References: Sterile compounding, radiopharmaceuticals, NatMed, geriatrics, pediatrics
6. Federal & Provincial Legislation, OCP Policies and Guidelines, By-Laws, Code of Ethics, Standards of Practice, etc.

For more information, visit:



RxTx was the name of an online resource that contained information from 3 print publications:

1. Compendium of Pharmaceuticals and Specialties (CPS)
2. Compendium of Therapeutic Choices (CTC)
3. Compendium of Therapeutics for Minor Ailments (CTMA)

Now, RxTx has been renamed to CPS and the 3 publications have been renamed, as well:

1. CPS → CPS: Drug Information
2. CTC → CPS: Therapeutic Choices
3. CTMA → CPS: Minor Ailments

The last print edition of CPS: Drug Information available was from 2022.

Consult CPS online, rather than the print editions, for the most up-to-date information.

SECONDARY REFERENCES: MEDLINE/PUBMED

MEDLINE®

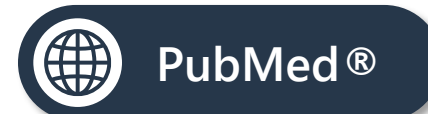
MEDLINE® is the National Library of Medicine® (NLM®) journal citation database with >29 million references to more than 5,600 scholarly journals.

Includes categorization of references to a controlled vocabulary called Medical Subject Headings (MeSH).



PubMed®

PubMed® is a website that allows you to access MEDLINE® plus additional references such as unapproved/uncategorized articles.

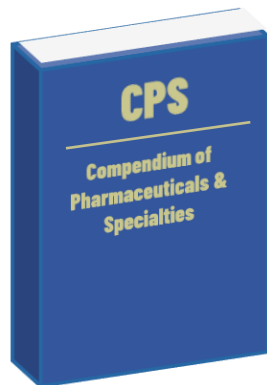


Use PubMed® or MEDLINE® when you want to search for primary literature

TERTIARY REFERENCES: DRUG PRODUCT MONOGRAPHS

Compendium of Pharmaceuticals & Specialties (CPS)

- Canadian
- Online resource for drug monographs
- Last print edition was in 2022



There are two types of monographs in CPS:

- **Product Monographs**
 - Provided, word-for-word, by pharmaceutical manufacturers and are approved by Health Canada
- **CPhA Monographs**
 - Numerous monographs compiled by editorial staff at the CPhA
 - Peer-reviewed by Canadian experts and provide evidence-based recommendations and clinical information about the drug/drug class
 - **Do not include information about physical/chemical properties and availability outside North America**

TERTIARY REFERENCES: DRUG PRODUCT MONOGRAPHS

Health Canada Drug Product Database

- Canadian
- **Drug monographs and availability (Canadian products)**
- Managed by Health Canada (federal regulator of therapeutic products)
- Does **NOT** provide medical advice on the use of products but contains drug specific information on drugs **APPROVED** for use in Canada



Health Canada | Drug Product Database

TERTIARY REFERENCES: DRUG INFORMATION

UpToDate® Lexidrug™ (formerly Lexicomp®)

- American
- **Drug interactions***
- Alternative to CPS:
 - Drug monographs
 - Drug identification
 - IV compatibility
 - Calculations
 - Patient education
 - Toxicology
 - Off-label uses
- Alternative to Pediatric and Neonatal Dosage Handbook
 - Pediatric dosing



*CPS uses Lexidrug drug interaction tool, called Lexi-Interact

Micromedex

- American
- Alternative to CPS:
 - Drug monographs
 - Drug identification
 - Drug interactions
 - IV compatibility
 - Calculations
 - Patient education
 - Toxicology
 - Off-label uses



TERTIARY REFERENCES: DISEASE/DRUG INFORMATION

UpToDate®

- American
- Evidenced-informed information on disease states
- Links to Lexidrug drug monographs



Merck Manual

- American
- Evidenced-informed information on disease states
- Drug Information via UpToDate® via Lexidrug™
- Free
- Has professional, consumer, veterinary version



TERTIARY REFERENCES: DRUG INFORMATION

Handbook of Clinical Drug Data

- American
- Pharmacokinetic and lab parameters monitoring



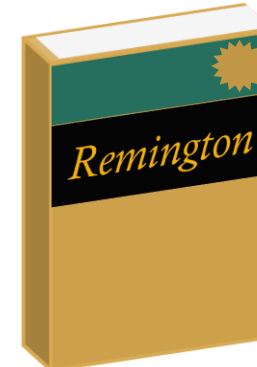
Martindale (The Complete Drug Reference)

- Published in UK
- Foreign drug search (all products around the world)



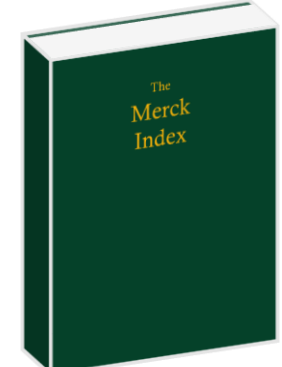
Remington (The Science and Practice of Pharmacy)

- American
- Basic information about compounding techniques and ingredients



Merck Index

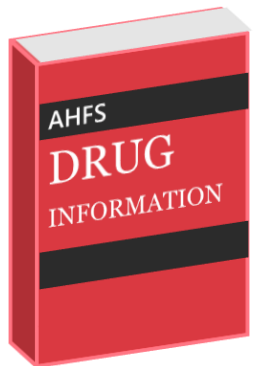
- American
- Useful for compounding
- Chemical and pharmacologic information about a variety of products



TERTIARY REFERENCES: DRUG INFORMATION

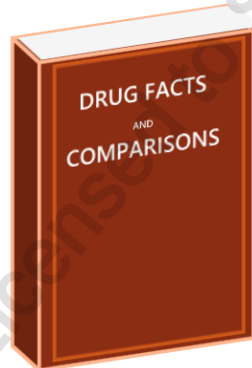
AHFS Drug Information

- American
- Monographs of American FDA-approved and off-label medications



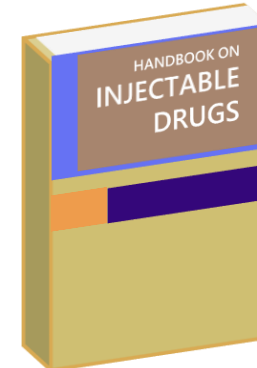
Drug Facts and Comparisons

- Off-label uses
- Product use, available dosage forms and sizes, and manufacturer information



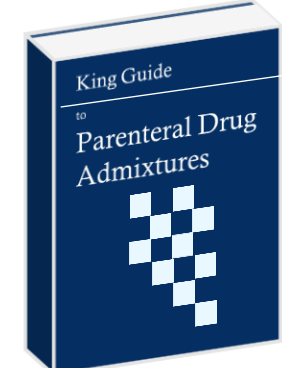
Handbook On Injectable Drugs (Trissel)

- American
- Hospital practice
- IV prep, stability and compatibility of parenteral medications



King Guide to Parenteral Drug Admixtures

- American
- Hospital practice
- IV prep, stability and compatibility of parenteral medications



TERTIARY REFERENCES: DRUG COMPARISONS

RxFiles

- Canadian
- Comprehensive summary of each disease state (recommendations, dosage forms, doses, pricing, etc.), **NOT drug monographs**
- **Preferred source for drug comparisons**
- Academic Detailing (AD) Program
 - Academic detailing: usually a one-on-one interaction involving trained health care professionals with clinicians. Provides clinician with evidence-based information in order to facilitate good clinical decision making. Provides high quality information



TERTIARY REFERENCES: THERAPEUTICS

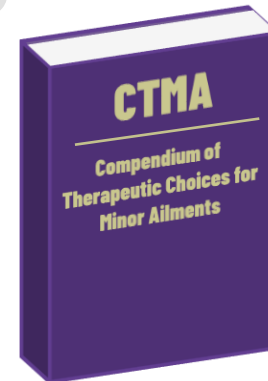
CPS: Therapeutic Choices (formerly Compendium of Therapeutic Choices)

- Canadian
- Therapeutic information (evidence-based)
- Contains:
 - Information on disease states
 - Algorithms and drug tables (drug comparisons)
 - Drug use in pregnancy and breastfeeding
 - Not comprehensive
 - Dosage adjustment in renal/hepatic impairment
 - **Might NOT contain the most up-to-date information, therefore, should refer to the clinical practice guidelines for the most up-to-date recommendations**

TERTIARY REFERENCES: SELF-CARE

CPS: Minor Ailments (formerly Compendium of Therapeutic Choices for Minor Ailments)

- Canadian
- Primarily self-care or minor ailments (evidence-based)
- Contains information on:
 - Pathophysiology
 - Goals of therapy
 - Non-pharmacological management
 - Pharmacological therapy
 - Some information on pregnancy
 - Some natural health products



TERTIARY REFERENCES: QT RISK ASSESSMENT

CredibleMeds

- American
- Evidence-based internationally-recognized database categorizing drugs by their potential to cause QT prolongation and/or torsades de pointes (TdP)
 - Includes drug lists, a searchable database and clinical guidance
 - Used to evaluate drug safety, particularly in patients with arrhythmias
- Accessible to Canadian healthcare practitioners for free following registration
- Maintained by the Arizona Center for Education and Research on Therapeutics (AZCERT)



CredibleMeds

TERTIARY REFERENCES: NATURAL HEALTH PRODUCTS (NHPs)

NatMed Database Therapeutic Research Center

- American
- Primarily NHPs
 - Drug interactions
 - Safety and efficacy
 - Nutrient depletion checker
 - Pregnancy and lactation
 - Adverse effects
- Search by disease OR product



NatMed Database

Licensed Natural Health Product Database (LNHPD) HEALTH CANADA

- Canadian
- Primarily licensed NHPs in Canada (and information on availability)
- Brief drug monographs



Licensed Natural Health Product Database

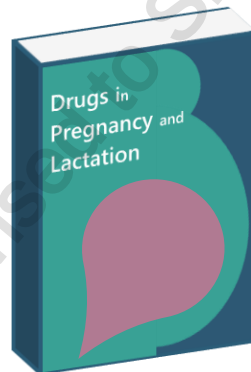
TERTIARY REFERENCES: PREGNANCY AND LACTATION

Drugs in Pregnancy and Lactation (Briggs)

- American
- Pregnancy and breastfeeding
 - Drug name and class (both prescription and non-prescription products)
 - Provides information on studies in pregnancy and lactation
 - Provides general recommendation on use in pregnancy and lactation



Use Briggs for
information on drug
safety in pregnancy and
lactation



Drug and Lactation Database (LactMed®)

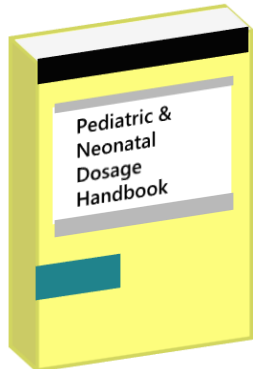
- American
- Drug use during breastfeeding only
- NOT comprehensive (does not contain information on many medications)



TERTIARY REFERENCES: PEDIATRIC & GERIATRICS

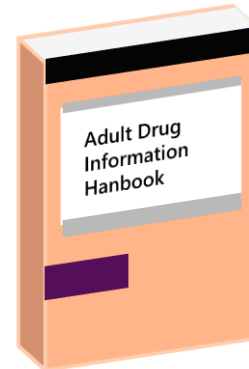
Pediatric & Neonatal Dosage Handbook

- American
- Published by UpToDate® Lexidrug™



Adult Drug Information Handbook

- American
- Published by UpToDate® Lexidrug™



TERTIARY REFERENCES: CONTINUING EDUCATION

- Provides pharmacists with current evidence-based information on disease states, ensuring competency throughout their career
- Offered by:
 - rxBriefCase = www.rxbriefcase.com (Need to know this)
 - Rogers publishing = www.canadianhealthcarenetwork.ca
 - Pear Healthcare = www.healthlearning.ca
 - PharmAchieve = www.pharmachieve.com/register#ces
 - Employer conferences
 - Live and local events
 - Association conferences
 - Multidisciplinary conferences
 - Provincial/Territorial College website (Need to know this)
 - Canadian Pharmacists Association (CPhA) (Need to know this)
 - Canadian Council on Continuing Education in Pharmacy (CCCEP)

TERTIARY REFERENCES: NATIONAL ORGANIZATIONS

REGULATORY

NAPRA (National Association of Pharmacy Regulatory Authorities)

- Association of pharmacy regulatory bodies
- Does not actually regulate the pharmacy profession (the provincial and territorial regulators do) but it serves as a place where the provincial and territorial regulators come together to harmonize regulations. This is accomplished through recommended standards which are developed by consensus and published by the NAPRA website
- These recommended standards are often closely or automatically adopted by the regulators. For instance, NAPRA's drug recommended drug schedules, which are published by its National Drug Scheduling Advisory Committee (NDSAC), effectively serves as the decider of drug schedules for 8 provinces/territories because those provinces automatically follow its recommendations
- Also publishes "Entry to Practice Competencies" which the PEBC® exams are based on

TERTIARY REFERENCES: NATIONAL ORGANIZATIONS

ADVOCACY

CPhA (Canadian Pharmacists Association)

- Largest organization of pharmacists in Canada
- **Advocates on behalf of pharmacists** to governments and the public (example: they have a campaign to expand the scope of practice of pharmacists)
- Leads a number of national projects, publications, conferences
- Lobbies governments and publishes recommended government policies
- **Publishes references like CPS: Drug Information, CPS: Therapeutic Choices, CPS: Minor Ailments**
- Actively enhances the reputation of the profession
- Also **offers CE** (Continuing Education) programs

TERTIARY REFERENCES: SAFETY

ISMP Canada (Institute for Safe Medication Practices Canada)

- <http://www.ismp-canada.org/index.htm>
- Reporting medication incidents (CMIRPS)
- Safety alerts
- Promotes safe medication practices (e.g. dangerous abbreviations)

Health Canada Advisories, Warnings, Recalls

- <http://www.hc-sc.gc.ca/ahc-asc/media/advisories-avis/index-eng.php>
- Comprehensive lists of recalls, advisories, and safety alerts

MedEffect Canada (Health Canada)

- <https://www.canada.ca/en/health-canada/services/drugs-health-products/medeffect-canada.html>
- Can submit adverse reaction reports (designed for both health professionals and consumers)
- Links to Health Canada Advisories, Warnings, Recalls

Canada Vigilance Program (Health Canada)

- <https://www.canada.ca/en/health-canada/services/drugs-health-products/medeffect-canada/canada-vigilance-program.html>
- Health Canada's post-market surveillance program
- Can search database for adverse effects

HOW DO YOU DECIDE WHICH REFERENCE TO USE?

**MUST KNOW
FOR EXAM**

1. Is the reference Canadian?

- All else being equal, always pick the Canadian source because non-Canadian sources may have unapproved indications / unapproved therapeutic uses, be based on different guidelines and have different units of measure.

2. Is the reference specific to my question(s)?

- E.g. you would use Briggs for drug safety in pregnancy and lactation over other sources because it is more specific and contains more information in that area.
- E.g. you would use a product monograph for information on how to administer a drug instead of PubMed.

3. Is it the most comprehensive?

- A more complete reference is usually preferred
- E.g. you would use Health Canada Drug Product database over CPS to find dosage forms because it contains all licensed products in Canada whereas CPS contains most, but not all, licensed products.

HOW DO YOU DECIDE WHICH REFERENCE TO USE?

**MUST KNOW
FOR EXAM**

4. Is it the right level of detail?

- E.g. you generally would NOT turn to PubMed to find out how to treat a disease state
- Instead, you would go to the latest consensus guidelines or CPS: Therapeutic Choices which has already distilled many (sometimes contradictory) studies into evidence-based recommendations.

5. Is it the most up to date?

- The online clinical practice guidelines are more up-to-date than the print version.
- The online edition of CPS: Therapeutic Choices is more up-to-date than the print edition.

6. Is it trusted?

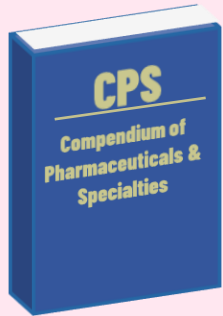
- Try and use peer-reviewed publications. You would not use Wikipedia.

TAKEAWAY

- **Primary references:** Original research authored by researchers and often published in peer-reviewed journals. Also includes conference proceedings, technical reports.
- **Secondary references:** Work that builds on original research like databases; meta-analyses that review original research and practice guidelines
- **Tertiary references:** Distillation and collection of primary and secondary resources like textbooks
- **Natural health products resources:** NatMed database, Health Canada Natural Health Product database
- **Pregnancy and Lactation resources:** Drugs in Pregnancy and Lactation (Briggs), Drug and Lactation database (LactMed®)
- **Compounding resources:** Remington, Merck Index
- **Drug interaction resources:** Micromedex, Lexi-Interact

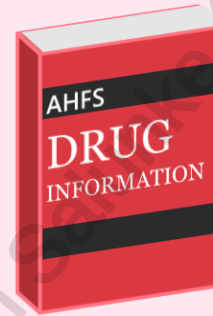
COMPARISON OF REFERENCES

DRUG MONOGRAPHS



CPS has CANADIAN:

- Brand Names
- Monographs
- Indications
- Approved Uses
- Units of Measure



AHFS has American:

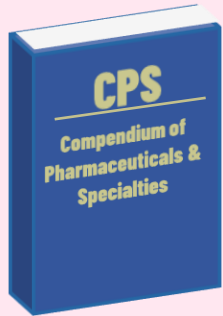
- Brand Names
- Monographs
- Indications
- Approved Uses
- Units of Measure



All else being equal, always **pick the Canadian source** because non-Canadian sources may have unapproved indications / unapproved therapeutic uses, be based on different guidelines and have different units of measure

COMPARISON OF REFERENCES

DRUG MONOGRAPHS



CPS has CANADIAN:

- Brand Names
- Monographs
- Indications
- Approved Uses
- Units of Measure



UpToDate® Lexidrug™
(formerly Lexicomp®) has
American:

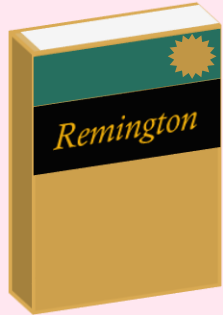
- Brand names (includes Canadian brand names)
- Monographs
- Drug identification
- Dosing schedules (includes Canadian dosing)



All else being equal, always **pick the Canadian source** because non-Canadian sources may have unapproved indications / unapproved therapeutic uses, be based on different guidelines and have different units of measure

COMPARISON OF REFERENCES:

COMPOUNDING



Remington

- Basic information about compounding techniques and ingredients



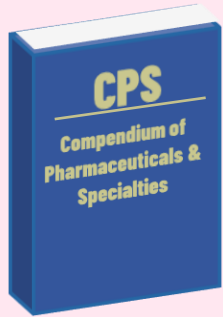
Merck Index

- Useful for compounding
- Chemical and pharmacologic information about a variety of products



COMPARISON OF REFERENCES

PREGNANCY AND BREASTFEEDING



CPS Canadian monographs contain information on pregnancy and lactation – not comprehensive

- Not first choice reference for pregnancy and lactation



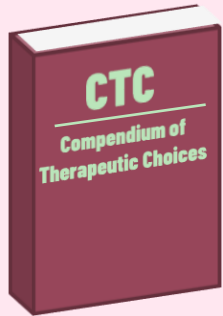
Briggs

- Primary resource
- More comprehensive compared to CPS



COMPARISON OF REFERENCES

THERAPEUTICS



CPS: Therapeutic Choices

- Might not contain the most up-to-date information, therefore, refer to the clinical practice guidelines

Clinical Practice Guidelines

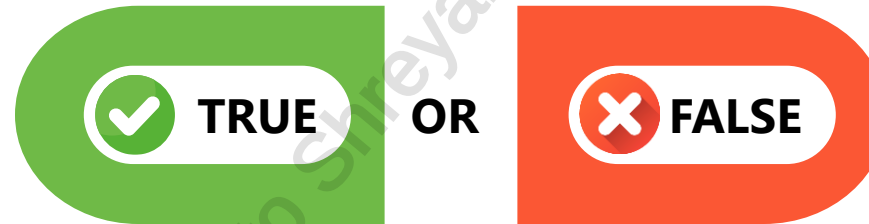


Clinical Practice Guidelines provide clinicians with the most up-to-date recommendations

COMPARISON OF REFERENCES

DRUG PRICING

You can find drug prices in RxFiles

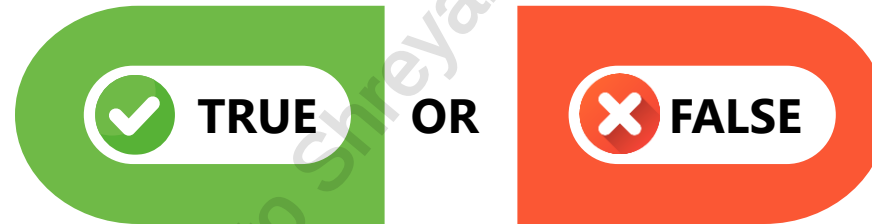


You can also find drug prices in CPS: Therapeutic Choices, however, the prices are shown as dollar signs.

COMPARISON OF REFERENCES

CE PROGRAMS

CE (Continuing Education) programs are offered by DIRC (Drug Information and Resource Center) and RxBriefCase

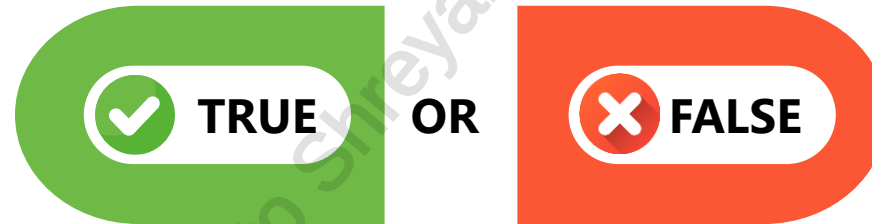


CE programs are offered by RxBriefCase, PharmAchieve, Provincial/Territorial College Websites, and CPhA. There are other ways to obtain CE credits, however, the ones mentioned are the most important. DIRC is a phone service that provides clinicians with answers to drug information inquiries.

COMPARISON OF REFERENCES

TREATMENTS

PubMed® is the best resource to find recommendations/treatment options for each disease state

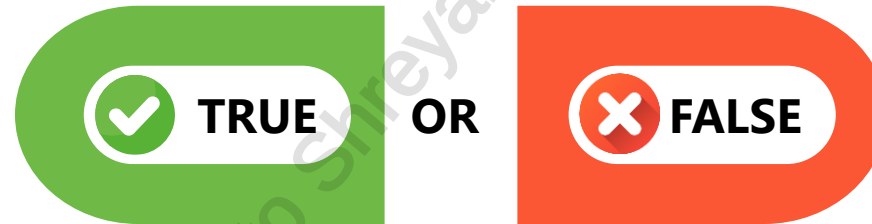


PubMed allows clinicians to find journal articles specific to certain patient populations. For example, assume that a doctor wants to know if taking 7 tablets of levothyroxine every week is as effective/safe as taking one tablet a day. For general inquiries about each disease state, please refer to guidelines or CPS: Therapeutic Choices.

COMPARISON OF REFERENCES

NHPS

The NatMed database can be used to determine if St. John's Wort interacts with paroxetine



PRACTICE QUESTIONS

1. GV is a 49-year-old patient who has recently moved from Austria. He comes to your pharmacy requesting a prescription for a drug called "Ospexin®". After consulting both Canadian and American drug references, you cannot find any information on this drug. Which of the following references is the most appropriate to consult next?
 - a) Merck Index
 - b) Martindale
 - c) CPS: Drug Information
 - d) Micromedex

2. You work in a compounding pharmacy and wish to learn more about the chemical and pharmacological properties of a compound you are using. Which of the following is the most appropriate resource to consult?
 - a) Lexi-Interact
 - b) UpToDate®
 - c) NatMed
 - d) Merck Index

REFERENCES

1. Additional Resources – OCPIInfo.com. (2018). <https://www.ocpinfo.com/regulations-standards/additional-resources/>
2. CPS Subscriptions - English. www.pharmacists.ca. (n.d.). <https://www.pharmacists.ca/products-services/cps-subscriptions/>
3. CPS Print - English. www.pharmacists.ca. (n.d.). <https://www.pharmacists.ca/products-services/cps-print/>

CHANGE LOG

July 2025

- Minor formatting changes made throughout
- Slide 7: Updated links

September 2025

- Slide 24: Updated link to MedEffect Canada
- Minor formatting changes made throughout

February 2026

- Slide 2: Added new NAPRA competencies
- Slide 10: Added 'off-label uses' to Micromedex, added name of CPS drug interaction tool